10

What Is the Cloud Future?

Without a doubt, the future of business computing (and personal computing) resides in cloud-based environments. The cloud is a relatively new technology and its full potential remains to be reached. Most businesses recognize the value in cloud computing and view it as a solution to many of their challenges–technical, managerial, and financial.

Aside from "more of the same" advantages the future of cloud computing offers, such as elasticity, monetary savings, reduction of on-premise data center expenses, and so forth, many exciting developments and new ideas emerge with cloud computing. Among these are NoOps, Zero Knowledge Clouds, Serverless Architectures, Machine Learning, Cloud Streaming Services, Edge computing, and others. We will look at several cloud-based trends emerging to help business leaders and cloud subscribers envision their future.

NoOps

NoOps is an abbreviation for *No Operations*. This idea suggests IT infrastructures in organizations could become so automated that operations personal would no longer be needed. As you might imagine, this idea sends chills up the spine of people with careers in IT operations and has caused more than a few Twitter storms over the past year.

The NoOps concept recognizes someone must set up the automation but once running, a dedicated IT operations staff is not needed. A more realistic way to describe this trend would be as: *more automation*.

Without a doubt, the cloud's future includes more automation, and intelligent software systems that abstract organizational needs from usage patterns and translate that into operational environments. Automation is useful for anomaly detection and understanding when resources are compromised or used for unintended purposes. In the near term, with the DevOps movement, and powerful automation and orchestration, fewer operations personnel will be required. Instead, new roles related to integrating secure cloud operations with business needs will emerge. In the more distant future, perhaps machine learning, AI, and other technologies will provide most IT operations services. However, organizational needs, SLAs, and other business-side processes will remain within the IT operations specialist domain for a long time to come.

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Companion website: www.wiley.com/go/mchaney/cloudtechnologies

Everything as a Service (EaaS)

As mentioned in Chapter 9, cloud-based application services have grown at an incredible rate. Practically everything in computing environments now runs in the cloud. This *EaaS* trend is not slowing. Most new activity relates to business application software and consumer products such as video and music streaming, and mobile apps.

Services that rely on subscription charges have been embraced by business and consumer alike. Netflix and other vendors demonstrate that steady cash flows appeal to investors. From a business perspective, costs are predictable and managed more easily in these environments which makes cloud-based services even more attractive.

Zero Knowledge Cloud Storage

Security is important regarding the cloud's future. With regular news about data breaches and cyber-criminals stealing millions of consumer records, this area must be considered for future enhancements.

Zero-knowledge cloud storage is an idea that makes cloud computing more secure. Although this has been available for some time, it suffers from speed and management issues which prevent some organizations from using this ultra-secure method of storing data. For many companies, only a small portion of their most sensitive data is stored this way.

So, what is zero-knowledge cloud storage? It simply means all data is encrypted prior to movement into the cloud so the cloud provider and anyone working in their data center cannot access the data in a meaningful or useful way. While it offers higher security, several challenges exist, and several future solutions will emerge to make zero-knowledge cloud storage a much wider practice. First, encryption keys in zero-knowledge storage are kept by the data owner only. Therefore, if the key is lost or corrupted, data cannot be retrieved in a useful way. If the key is lost, the data is gone for all practical purposes.

The other downside relates to storage and transmission speeds. Since encryption and decryption operations take place prior to cloud storage, operations are slowed. This makes real time use of this technology unviable in many applications. Cloud vendors hope to speed up zero-knowledge cloud storage as well as find better approaches to accomplish the same outcome–highly secure data storage. Without a doubt many new developments will emerge in this area.

Serverless Architecture

Serverless architecture is a computing model receiving renewed attention among large cloud providers (e.g. Google, Amazon, and Microsoft), and third-party vendors hoping to develop a new market. Serverless technology is consistent with the idea of NoOps, where cloud clients do not worry about day-to-day operations in their data centers. Google already uses

similar technology to power its largest applications like Gmail and YouTube. Serverless architecture takes the concept of spinning up new servers when demand requires (or deprovisioning underutilized servers) and automates it. So, serverless architecture refers to the concept of dynamically automating the level of infrastructure needed and then provisioning (or deprovisioning) the resources required in real-time to keep operations running optimally.

The concept does have challenges such as latency (e.g. the delay between resource need and resource availability), security, and privacy. Special care must ensure remnants of deprovisioned servers and critical data are not languishing in cloud memory. However, with DevOps gaining popularity, using serverless architecture appears to be gaining momentum.

Multicloud

Large organizations often rely on business models using resources from multiple vendors. For instance, the best-in-class application software may come from one source (e.g. Salesforce or SAP), productivity software from another (Office 365), and so forth. This means an organization may already deal with multiple public clouds in its IT infrastructures. In addition, many organizations hesitate to rely on a single cloud provider for all needs and do not wish to put all "eggs in a single basket." *Multicloud* is not the same as a hybrid cloud which relies on using a combination of on-premise, private, and public clouds. Instead, multicloud engages multiple cloud vendors. The motivation for going with a multicloud approach is complex. It can reduce cloud subscriber reliance on one vendor to provide better leverage when negotiating future SLAs. It could alleviate governance concerns and ensure business needs are met. It might increase computing or communication speed due to geographic proximity or because one mission critical application requires higher speed and it is too expensive to deploy the entire organizational computing platform that way. In short, the idea of picking and choosing clouds from a variety of sources can add business value and will undoubtably continue into the foreseeable future.

Small Business Clouds

Cloud computing continues to grow in importance for small and medium enterprises (SMEs). This is true particularly for start-ups and new cash-strapped innovations seeking to minimize capital expenditures. In fact, many new cloud computing models such as NoOps and subscription-based services ensure costs are based on consumption and computing services easily scale up as an enterprise grows. Many SMEs lack the ability to hire and maintain an IT staff and using cloud computing with data center services can replace this need in the short term, and in some cases even for the long term. Top-notch help desk and IT specialists are available through cloud vendors. Also, a SME can extend its reach into global markets using cloud-based IT services.

Machine Learning

Machine learning (ML) is an exciting area in both cloud computing and the broader IT world. In general terms, ML involves development of software that becomes more accurate in predicting outcomes without the intervention of humans. In other words, algorithms in ML find better ways of achieving goals and improve over time.

Several different approaches are used in ML including supervised and unsupervised learning. Supervised learning includes a more significant human dimension in the process where data scientists provide feedback to the model as it develops and determine what variables should influence the learning behavior. Supervised learning involves a training phase prior to use on a full data set. See Figure 10.1.

Unsupervised learning, on the other hand, uses algorithms and iterative cycles to explore data and arrive at recommendations. Often, unsupervised learning involves neural networks and solves complex problems in domains such as speech recognition, image classification, and those that involve rich media.

ML provides exciting possibilities for cloud computing and fits nicely into the mix with DevOps, NoOps, and serverless computing architectures. For example, in the IaaS arena, longitudinal data can determine the best time to provision a new VM, so no latency issues are experienced by users. Likewise, patterns of resource use can inform an ML algorithm whether a security breach is likely or not. On the SaaS side, applications can use ML to look at email patterns to determine if a customer is likely to leave for a competitor, or ML can help determine the best time to purchase goods and services in a manufacturing operation. ML also can be useful in powering help desk applications. The uses of ML are nearly endless as this technology matures and moves into the mainstream.

Major cloud vendors all offer ML packages for inexpensive deployment. These packages interface with cloud-based data collection and provide insights to help identify issues, predict needs, or discover ways of economizing. The packages generally are deployed as SDKs and APIs for use with application software and database systems. Using ML currently requires enterprise expertise and may necessitate employing a data scientist or business analyst.

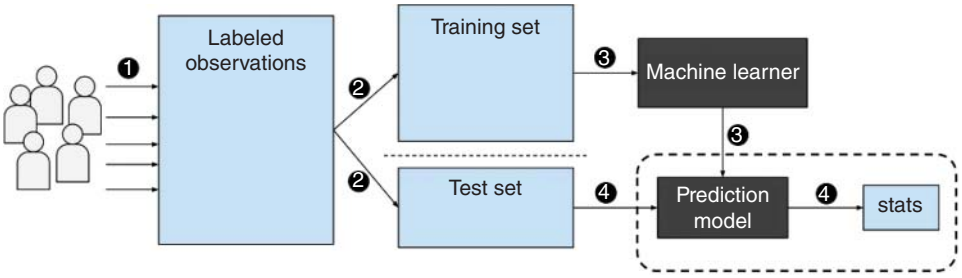

Figure 10.1 Supervised machine learning process. Source: EpochFail, Supervised machine learning process.

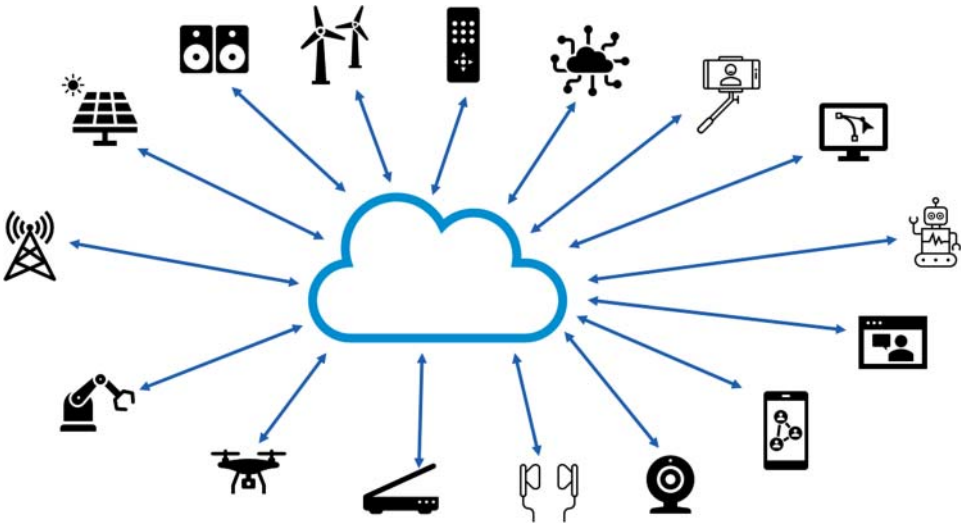

Figure 10.2 IoT pulls data from a wide range of sources for processing in the cloud. Source: Pixabay, IoT pulls data from a wide range.

Internet of Things (IoT)

Cloud computing is a natural fit with the *Internet of Things (IoT)*. Large data sets are pulled into the cloud from a wide variety of devices. This is compatible with cloud attributes such as scalable storage, variable processing power, and analysis tools. Devices ranging from personal fitness bands to home assistants like Amazon's Echo to GPS devices tracking automobile movement provide rich data sources for organizations seeking to leverage ML and other techniques to improve operations or more effectively interact with customers and business processes. Additional data comes from social media, sensors, and mobile devices. The cloud provides required computing power and storage capacities while IOT offers new ways to pull data from everyday devices in ways that seem like something out of science fiction. This technology provides a glimpse of its future business potential and plays a key role in organizational planning. Figure 10.2 illustrates.

Cloud Computing as a Utility

Some experts foresee cloud computing moving to a greater utility-type model as time goes on–both for private individuals and organizations. In many ways, cloud computing already operates as a utility, but will become more so with automatic provisioning and service option choices. The model is like electric utilities or city water metering. As computing resources are consumed, a monthly fee is assessed and paid by the user. This approach promises that shared resources reduce costs and improve power for everyone in the long run.

Cloud Streaming Services

Cloud streaming services are set for big changes and functional improvements. Consumer demand for streaming services has increased rapidly and expectations for entertainment videos, music, television channels, virtual reality applications, video games, online courses, digital books, and other material continues to increase. Businesses require content servers for training material, organizational resources, customer help material, and digitalized records. Individuals want to play games, watch movies, listen to music, read books, and interact with peers using cloud-based systems. These areas are certain to continue growing and placing demands on network bandwidth and storage resources.

Edge Computing

Edge computing promises to add a layer of computing power that enhances the nexus between cloud computing and IoT. In its most basic form, Edge computing relies on technologies that allow computation to occur on a network's edge–usually on downstream cloud data and upstream IoT data (Shi et al. 2016). See Figure 10.3. Basically, essential data processing occurs closer to IoT devices or other sources via a mobile, Bluetooth, or wireless

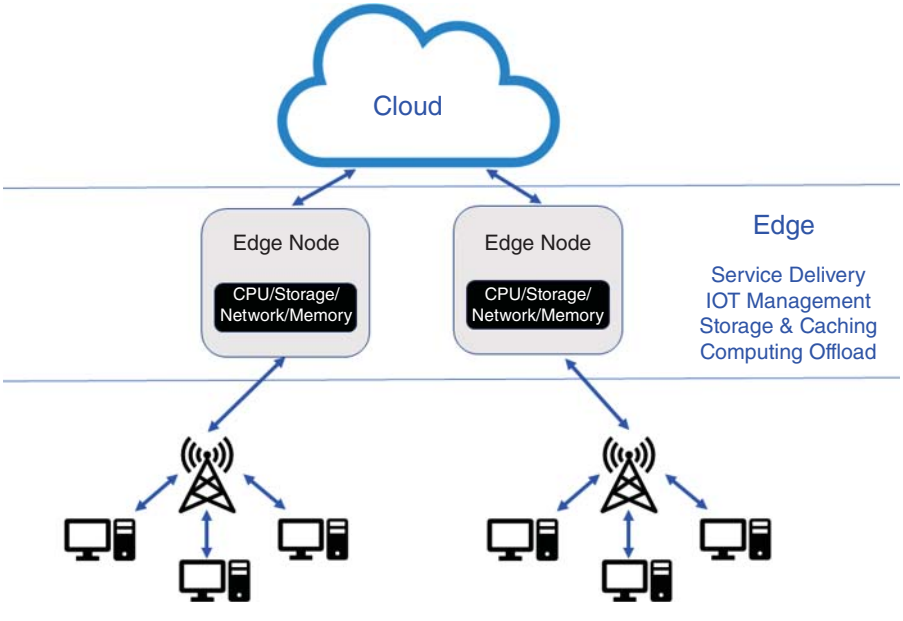

Figure 10.3 Edge computing architecture example. Source: NoMore201, Edge computing architecture example.

network. According to De Donno et al. (2019), Edge computing is implemented in various ways. Among these are *Mobile Cloud Computing (MCC)* and *Cloudlet Computing (CC),* or *Mobile Edge Computing (MEC)*. At its core, MCC advocates mobile devices offload storage and computation to reduce the processing load. This idea suggests the best candidates for processing and storage are devices at a network's edge. This enables pre-processing prior to data moving into a broader cloud environment. CC is a leading implementation of MCC to enable IoT's integration with the cloud. According to De Donno et al. (2019), "[a] cloudlet is a trusted, small Cloud infrastructure, located at the edge of the network and available for nearby mobile devices that collaborates with the Cloud to compute the results and then sends them back to mobile devices." Another approach to Edge computing is MEC. This brings computation and storage to a mobile network's edge via a Radio Access Network. This offers context awareness, scalability, location responsiveness, low latency, and high bandwidth potential.

Fog Computing

Fog computing takes Edge to a higher level. It offers distributed computing, storage, control, and networking capabilities closer to the user (Chiang et al. 2017). According to De Donno et al. (2019) this is:

> *the highest evolution of the Edge computing principles. Indeed, Fog computing is not limited to only the edge of the network, but it incorporates the Edge computing concept, providing a structured intermediate layer that fully bridges the gap between IoT and Cloud computing. In fact, Fog nodes can be located anywhere between end devices and the Cloud, thus, they are not always directly connected to end devices. Moreover, Fog computing does not only focus on the "things" side, but it also provides its services to the Cloud. In this vision, Fog computing is not only an extension of the Cloud to the edge of the network, nor a replacement for the Cloud itself, rather a new entity working between Cloud and IoT to fully support and improve their interaction, integrating IoT, Edge and Cloud computing.*
>
> (p. 150936)

Fog computing utilizes three-tier architecture as shown in Figure 10.4. The IoT Layer represents geographically dispersed devices, smart appliances, smart homes, drones, industrial sensors, crop moisture sensors, or any other IoT device.This interacts with the Fog layer where Fog nodes compute, communicate, and store data temporarily. Finally, the Cloud layer receives processed data into its centralized infrastructure where PaaS or IaaS processes integrate, store, and use the data. Fog provides several benefits to the interface between IoT and Cloud Computing. These advantages have been called SCALE: Security, Cognition, Agility, Latency, Efficiency and represent how preprocessing closer to the user or device improves cloud operations (Chiang et al. 2017; De Donno et al. 2019).

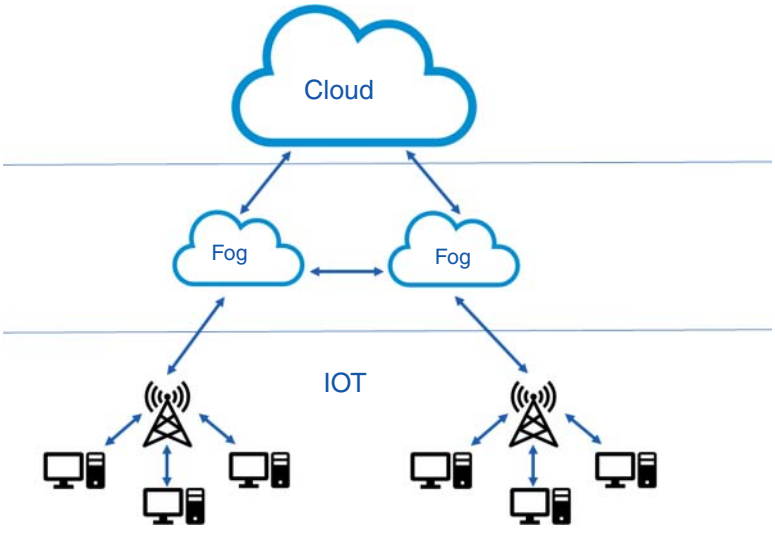

Figure 10.4 Three-tier Fog computing architecture example. Source: Michele De Donno, Three-tier Fog computing architecture example.

Summary

Commercial cloud computing has existed for about a decade and has dramatically altered organizational approaches to IT infrastructure and application delivery. Cloud computing has triggered enormous innovation, new products, and ideas. The very thought that data centers could become virtual entities located thousands of miles from user bases is incredible, yet it has happened everywhere.

The next generation of cloud computing shows no signs of slowing down or stagnating. Technologies like machine learning, serverless architecture, AI, and IoT promise advances even more incredible than what we have today. The convergence of organizational and private computing demands has triggered innovations in streaming technologies, virtualization, and Internet speeds. New ways to manage and control costs produce innovative SLAs between companies and cloud providers that offer stability, security, and frameworks for sound governance. Management of IT has integrated development and operations activities with DevOps and NoOps philosophies. This breaks down barriers and offers new ways to enhance organizational effectiveness.

Additional functions are virtualized and moved into elastic environments responding to needs rapidly and in ways that inhibit waste. Advanced monitoring systems collect data and interact with ML algorithms to prevent problems, security breaches, and operating faults. Edge computing moves processing closer to the source making IoT faster and better. Business analytics provide reports to decision makers and increasingly are enhanced with AI-based recommendations.

All in all, the cloud's future is bright and will impact EVERYONE but especially those in IT related fields (Figure 10.5).

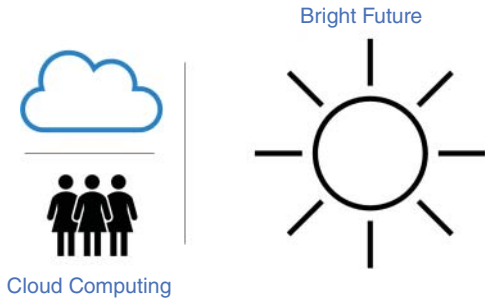

Figure 10.5 Future of cloud computing is bright!. Source: Public domain files. Future of cloud computing is bright. http://www.publicdomainfiles.com/show_file.php?id=13534699217949.

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